

```
In[*]:= << mGRG`STensor`
```

Chap. 5 Differential Form Operations

5.1 Basic Differential Form Operations

5.1.1 Defining Forms and Taking Their Components

2-form을 정의한다:

```
In[*]:= Fdefine[w2, 2]
```

0-form, 1-form, 3-form을 정의한다:

```
In[*]:= Fdefine[f, 0]; Fdefine[w1, 1]; Fdefine[w3, 3]
{f, w1, w2, w3}
```

Out[*]=

```
{f, w1, w2, w3}
```

0-form인가를 확인한다:

```
In[*]:= ZeroDegreeQ /@ %
```

Out[*]=

```
{True, False, False, False}
```

미분 형식(Differential Form)의 degree 또는 랭크를 얻는다:

```
In[*]:= DegreeForm /@ %%
```

Out[*]=

```
{0, 1, 2, 3}
```

미분 형식을 IndexedTensor로 변환한다:

```
In[*]:= FtoC[w3, {la, lb, lc}]
```

Out[*]=

```
w3abc
```

IndexedTensor 표기에서 미분 형식은 완전 반대칭 텐서이다.

```
In[*]:= {w3[lb, la, lc], w2[lb, la]}
% // TindexSort
```

Out[*]=

```
{w3bac, w2ba}
```

Out[*]=

```
{-w3abc, -w2ab}
```

```
In[*]:= ZeroDegreeQ[w3[la, lb, lc]]
DegreeForm /@ {w3[la, lb, lc], w3}
```

```
Out[*]= True
```

```
Out[*]= {0, 3}
```

미분 형식의 좌표계 표현을 얻기 위해 시공간을 3차원으로 설정하고, Indexed 0-form을 정의한다:

```
In[*]:= SetDimension[3];
Fdefine[X[ua], 0]
```

2-form w2의 좌표 표현을 얻는 순서이다:

```
In[*]:= FtoC[w2, {la, lb}]
1
- FtoC[w2, {la, lb}] × XD[X[ua]] ∧ XD[X[ub]]
2
```

```
Out[*]= w2ab
```

```
Out[*]= 1
- w2ab dXa ∧ dXb
2
```

```
In[*]:= % // SumDum
% // TindexSort
```

```
Out[*]= -1/2 w221 dX1 ∧ dX2 + 1/2 w212 dX1 ∧ dX2 - 1/2 w231 dX1 ∧ dX3 +
1/2 w213 dX1 ∧ dX3 - 1/2 w232 dX2 ∧ dX3 + 1/2 w223 dX2 ∧ dX3
w212 dX1 ∧ dX2 + w213 dX1 ∧ dX3 + w223 dX2 ∧ dX3
```

위의 순서를 한 번에 마치는 함수 'CordRep'를 호출한다:

```
In[*]:= CoordRep[w2, {X[1], X[2], X[3]}]
```

```
Out[*]= w212 dX1 ∧ dX2 + w213 dX1 ∧ dX3 + w223 dX2 ∧ dX3
```

원통 좌표계로 설정하자:

```
In[*]:= SetComponents[X[ua], {r, θ, z}]
```

3-form w3의 좌표 표현이다:

```
In[*]:=
1
3! FtoC[w3, {la, lb, lc}] × XD[X[ua]] ^ XD[X[ub]] ^ XD[X[uc]]
% // SumDum
% // TindexSort
```

Out[*]=

$$\frac{1}{6} w_{3abc} dx^a \wedge dx^b \wedge dx^c$$

Out[*]=

$$\begin{aligned} & \frac{1}{6} w_{321} dr \wedge dz \wedge d\theta - \frac{1}{6} w_{312} dr \wedge dz \wedge d\theta - \frac{1}{6} w_{231} dr \wedge dz \wedge d\theta + \\ & \frac{1}{6} w_{213} dr \wedge dz \wedge d\theta + \frac{1}{6} w_{132} dr \wedge dz \wedge d\theta - \frac{1}{6} w_{123} dr \wedge dz \wedge d\theta \end{aligned}$$

Out[*]=

$$-w_{123} dr \wedge dz \wedge d\theta$$

```
In[*]:= CoordRep[w3, {r, \theta, z}]
```

Out[*]=

$$-w_{123} dr \wedge dz \wedge d\theta$$

```
In[*]:= CoordRep[w3]
```

Out[*]=

$$w_{123} dx^1 \wedge dx^2 \wedge dx^3$$

5.1.2 The Exterior Product

```
In[*]:= ClearDimension[] (* default *)
```

미분 형식들 사이의 외적(Exterior Product)을 구한다:

```
In[*]:= XP[f, w1]
```

Out[*]=

$$f w_1$$

```
In[*]:= w1 ^ w3
```

Out[*]=

$$w_1 \wedge w_3$$

외적의 순서가 자동적으로 조정된다:

```
In[*]:= w3 ^ w1
```

Out[*]=

$$-w_1 \wedge w_3$$

랭크가 홀수인 경우 자기 자신과의 외적은 0이다:

In[*]:= {w2 ^ w1, w2 ^ w2, w3 ^ w3, w1 ^ w1}

Out[*]=

{w1^w2, w2^w2, 0, 0}

외적의 인자 개수는 제한이 없다:

In[*]:= w3 ^ w2 ^ w1 ^ w2 ^ w2

Out[*]=

-w1^w2^w2^w2^w3

미분 형식의 IndexedTensor 표현은 완전 반대칭 텐서이다. 따라서 랭크가 큰 표현의 FtoC 함수의 결과는 연산도 오래 걸리고, 출력 결과도 좋지 않다:

In[*]:= FtoC[w1 ^ w2, {la, lb, lc}]

Out[*]=

w1_c w2_ab - w1_b w2_ac + w1_a w2_bc

In[*]:= FtoC[w1 ^ w2 ^ w3, {la, lb, lc, ld, le, lf}]

Out[*]=

-w1_f w2_de w3_abc + w1_e w2_df w3_abc - w1_d w2_ef w3_abc + w1_f w2_ce w3_abd - w1_e w2_cf w3_abd +
w1_c w2_ef w3_abd - w1_f w2_cd w3_abe + w1_d w2_cf w3_abe - w1_c w2_df w3_abe + w1_e w2_cd w3_abf -
w1_d w2_ce w3_abf + w1_c w2_de w3_abf - w1_f w2_be w3_acd + w1_e w2_bf w3_acd - w1_b w2_ef w3_acd +
w1_f w2_bd w3_ace - w1_d w2_bf w3_ace + w1_b w2_df w3_ace - w1_e w2_bd w3_acf + w1_d w2_be w3_acf -
w1_b w2_de w3_acf - w1_f w2_bc w3_ade + w1_c w2_bf w3_ade - w1_b w2_cf w3_ade + w1_e w2_bc w3_adf -
w1_c w2_be w3_adf + w1_b w2_ce w3_adf - w1_d w2_bc w3_aef + w1_c w2_bd w3_aef - w1_b w2_cd w3_aef +
w1_f w2_ae w3_bcd - w1_e w2_af w3_bcd + w1_a w2_ef w3_bcd - w1_f w2_ad w3_bce + w1_d w2_af w3_bce -
w1_a w2_df w3_bce + w1_e w2_ad w3_bcf - w1_d w2_ae w3_bcf + w1_a w2_de w3_bcf + w1_f w2_ac w3_bde -
w1_c w2_af w3_bde + w1_a w2_cf w3_bde - w1_e w2_ac w3_bdf + w1_c w2_ae w3_bdf - w1_a w2_ce w3_bdf +
w1_d w2_ac w3_bef - w1_c w2_ad w3_bef + w1_a w2_cd w3_bef - w1_f w2_ab w3_cde + w1_b w2_af w3_cde -
w1_a w2_bf w3_cde + w1_e w2_ab w3_cdf - w1_b w2_ae w3_cdf + w1_a w2_be w3_cdf - w1_d w2_ab w3_cdf +
w1_b w2_ad w3_cdf - w1_a w2_bd w3_cdf + w1_c w2_ab w3_def - w1_b w2_ac w3_def + w1_a w2_bc w3_def

시공간의 차원을 설정한다:

In[*]:= SetDimension[3]

시공간의 차원보다 랭크가 큰 미분 형식은 0이다:

In[*]:= {w2 ^ w2, w1 ^ w3}

Out[*]=

{0, 0}

3차원 시공간에서 최대 랭크는 3이다:

In[*]:= **XP[w1, w2]**

$$\frac{1}{3!} \text{FtoC}[\%, \{\text{la}, \text{lb}, \text{lc}\}] \times \text{XD}[\text{X}[\text{ua}]] \wedge \text{XD}[\text{X}[\text{ub}]] \wedge \text{XD}[\text{X}[\text{uc}]]$$

Out[*]=
w1^w2

Out[*]=

$$\frac{1}{2} w_{1a} w_{2bc} \mathbf{dx}^a \wedge \mathbf{dx}^b \wedge \mathbf{dx}^c$$

In[*]:= % // SumDum
 % // TindexSort
 % // CollectForm

Out[*]=

$$\begin{aligned} & \frac{1}{2} w_{11} w_{232} \mathbf{dr} \wedge \mathbf{dz} \wedge \mathbf{d\theta} - \frac{1}{2} w_{12} w_{231} \mathbf{dr} \wedge \mathbf{dz} \wedge \mathbf{d\theta} - \frac{1}{2} w_{11} w_{223} \mathbf{dr} \wedge \mathbf{dz} \wedge \mathbf{d\theta} + \\ & \frac{1}{2} w_{13} w_{221} \mathbf{dr} \wedge \mathbf{dz} \wedge \mathbf{d\theta} + \frac{1}{2} w_{12} w_{213} \mathbf{dr} \wedge \mathbf{dz} \wedge \mathbf{d\theta} - \frac{1}{2} w_{13} w_{212} \mathbf{dr} \wedge \mathbf{dz} \wedge \mathbf{d\theta} \end{aligned}$$

Out[*]=

$$-w_{11} w_{223} \mathbf{dr} \wedge \mathbf{dz} \wedge \mathbf{d\theta} + w_{12} w_{213} \mathbf{dr} \wedge \mathbf{dz} \wedge \mathbf{d\theta} - w_{13} w_{212} \mathbf{dr} \wedge \mathbf{dz} \wedge \mathbf{d\theta}$$

Out[*]=

$$(-w_{11} w_{223} + w_{12} w_{213} - w_{13} w_{212}) \mathbf{dr} \wedge \mathbf{dz} \wedge \mathbf{d\theta}$$

In[*]:= **CoordRep[XP[w1, w2], {r, \theta, z}]**

Out[*]=

$$(-w_{11} w_{223} + w_{12} w_{213} - w_{13} w_{212}) \mathbf{dr} \wedge \mathbf{dz} \wedge \mathbf{d\theta}$$

In[*]:= **CoordRep[XP[w1, w2]]**

Out[*]=

$$(w_{11} w_{223} - w_{12} w_{213} + w_{13} w_{212}) \mathbf{dx}^1 \wedge \mathbf{dx}^2 \wedge \mathbf{dx}^3$$

5.1.3 The Exterior Derivative

랭크 p 인 미분 형식의 외미분(Exterior Derivative)은 랭크가 $p+1$ 인 미분 형식이다.

```
In[*]:= XD[f]
FtoC[%, {la}]
% // CDtoBD
```

```
Out[*]:= df
```

```
Out[*]:=  $\nabla_a \mathbf{f}$ 
```

```
Out[*]:=  $\partial_a \mathbf{f}$ 
```

```
In[*]:= expr = XD[w1]
FtoC[%, {la, lb}]
% // CDtoBD
% // TindexSort
```

```
Out[*]:= dw1
```

```
Out[*]:=  $\nabla_a w1_b - \nabla_b w1_a$ 
```

```
Out[*]:=  $\partial_a w1_b - \partial_b w1_a - \Gamma_{ab}^c w1_c + \Gamma_{ba}^c w1_c$ 
```

```
Out[*]:=  $\partial_a w1_b - \partial_b w1_a$ 
```

외미분을 IndexedTensor로 변환할 때 (torsion-free) 공변 도함수 대신 보통 도함수를 사용하도록 설정한다:

```
In[*]:= Off[XDtoCDfrag]
```

```
In[*]:= FtoC[expr, {la, lb}]
```

```
Out[*]:=  $\partial_a w1_b - \partial_b w1_a$ 
```

```
In[*]:= XD[w2]
FtoC[%, {la, lb, lc}]
```

```
Out[*]:= dw2
```

```
Out[*]:=  $\partial_a w2_{bc} - \partial_b w2_{ac} + \partial_c w2_{ab}$ 
```

시공간을 2차원으로 설정한다:

```
In[*]:= SetDimension[2]
```

성분이 각각 u, v 인 1-form의 좌표 표현이다:

```
In[*]:= oneform = u XD[x] + v XD[y]
```

```
Out[*]=  
u dx + v dy
```

```
In[*]:= XDoneform = XD[oneform]
```

```
Out[*]=  
du^dx + dv^dy
```

1-form du와 dv의 좌표 표현이다:

```
In[*]:= XD[u]^ = D[u[x, y], x] XD[x] + D[u[x, y], y] XD[y]
```

```
Out[*]=  
dy u^{(0,1)}[x, y] + dx u^{(1,0)}[x, y]
```

```
In[*]:= XD[v]^ = D[v[x, y], x] XD[x] + D[v[x, y], y] XD[y]
```

```
Out[*]=  
dy v^{(0,1)}[x, y] + dx v^{(1,0)}[x, y]
```

```
In[*]:= XDoneform // Factor
```

```
Out[*]=  
-dx^dy (u^{(0,1)}[x, y] - v^{(1,0)}[x, y])
```

2차원에서 2-form의 외미분은 0이다:

```
In[*]:= twoform = u XD[x] ^ XD[y]  
XD[twoform]
```

```
Out[*]=  
u dx^dy
```

```
Out[*]=  
0
```

du^dv의 좌표 표현이다:

```
In[*]:= XD[u]^ ^ XD[v]^  
% // Factor
```

```
Out[*]=  
dx^dy v^{(0,1)}[x, y] u^{(1,0)}[x, y] - dx^dy u^{(0,1)}[x, y] v^{(1,0)}[x, y]
```

```
Out[*]=  
dx^dy (v^{(0,1)}[x, y] u^{(1,0)}[x, y] - u^{(0,1)}[x, y] v^{(1,0)}[x, y])
```

동일한 연산을 미분 형식의 좌표 표현으로 직접 구하자:

```
In[*]:= SetComponents[X[ua], {x, y}]
```

```
In[*]:= XD[s] ^ XD[t]
          1
          --- FtoC[%, {la, lb}] × XD[X[ua]] ^ XD[X[ub]]
          2!
          % // SumDum
          % // CollectForm
```

```
Out[*]= ds^dt
```

```
Out[*]= -1/2 ∂at ∂bs dxa ∧ dxb + 1/2 ∂as ∂bt dxa ∧ dxb
```

```
Out[*]= ∂2t ∂1s dx ∧ dy - ∂2s ∂1t dx ∧ dy
```

```
Out[*]= (∂2t ∂1s - ∂2s ∂1t) dx ∧ dy
```

```
In[*]:= CoordRep[XD[s] ^ XD[t], {x, y}]
```

```
Out[*]= (∂2t ∂1s - ∂2s ∂1t) dx ∧ dy
```

$dr \wedge d\theta$ 를 $dx \wedge dy$ 로 표현하자:

```
In[*]:= On[EvaluateBDFlag]
```

```
In[*]:= SetCoordinates[{x, y}];
          r = Sqrt[x^2 + y^2]; θ = ArcTan[x, y];
```



```

In[*]:=

$$\frac{1}{2!} \text{FtoC}[XP[XD[r], XD[\theta]], \{la, lb\}] \times XD[X[ua]] \wedge XD[X[ub]]$$

% // SumDum
% // CollectForm
% // Simplify

```

Out[*]=

$$-\frac{x \partial_a \text{ArcTan}[x, y] \partial_b x \, dX^a \wedge dX^b}{2 \sqrt{x^2 + y^2}} - \frac{y \partial_a \text{ArcTan}[x, y] \partial_b y \, dX^a \wedge dX^b}{2 \sqrt{x^2 + y^2}} +$$

$$\frac{x \partial_a x \partial_b \text{ArcTan}[x, y] \, dX^a \wedge dX^b}{2 \sqrt{x^2 + y^2}} + \frac{y \partial_a y \partial_b \text{ArcTan}[x, y] \, dX^a \wedge dX^b}{2 \sqrt{x^2 + y^2}}$$

Out[*]=

$$\frac{x^2 \, dx \wedge dy}{(x^2 + y^2)^{3/2}} + \frac{y^2 \, dx \wedge dy}{(x^2 + y^2)^{3/2}}$$

Out[*]=

$$\left(\frac{x^2}{(x^2 + y^2)^{3/2}} + \frac{y^2}{(x^2 + y^2)^{3/2}} \right) dx \wedge dy$$

Out[*]=

$$\frac{dx \wedge dy}{\sqrt{x^2 + y^2}}$$

```

In[*]:= CoordRep[XP[XD[r], XD[θ]], {x, y}] // Simplify

```

Out[*]=

$$\frac{dx \wedge dy}{\sqrt{x^2 + y^2}}$$

```

In[*]:= r = .; θ = .; (*default*)

```

dx ∧ dy를 dr ∧ dθ로 표현하자:

```

In[*]:= SetCoordinates[{r, θ}]

```

```

In[*]:= SetComponents[X[ua], {r, θ}];
xx = r Cos[θ]; yy = r Sin[θ];

```

```

In[*]:= aRule = {XD[Sin[θ]] → Cos[θ] XD[θ], XD[Cos[θ]] → -Sin[θ] XD[θ]};

```

```
In[*]:= XD[xx] ^ XD[yy]
% /. aRule
% // Simplify
```

```
Out[*]:= -r Sin[θ] dr ^ dCos[θ] + r Cos[θ] dr ^ dSin[θ] + r^2 dCos[θ] ^ dSin[θ]
```

```
Out[*]:= r Cos[θ]^2 dr ^ dθ + r Sin[θ]^2 dr ^ dθ
```

```
Out[*]:= r dr ^ dθ
```

```
In[*]:= ClearDimension[] (* default *)
```

임의의 미분 형식 ω 의 $d^2 \omega$ 는 0이다:

```
In[*]:= XP[w1, w2, w3]
% // XD
% // XD
```

```
Out[*]:= w1 ^ w2 ^ w3
```

```
Out[*]:= -w1 ^ w2 ^ dw3 - w1 ^ dw2 ^ w3 + dw1 ^ w2 ^ w3
```

```
Out[*]:= 0
```

```
In[*]:= On[XDtoCDfrag] (*default*)
```

5.2 Advanced Differential Form Operations

5.2.1 The Lie Derivative

```
In[*]:= Tdefine[v[ua]]; Fdefine[w, 2]
```

벡터 v 에 의한 2-form w 의 Lie 도함수:

```
In[*]:= lie = LD[v, w]
```

```
Out[*]:=  $\mathcal{L}_v w$ 
```

공식:

$$\mathcal{L}_v w_{ab} = v^c \nabla_c w_{ab} + (\nabla_a v^c) w_{cb} + (\nabla_b v^c) w_{ac}$$

```
In[*]:= FtoC[lie, {la, lb}]
```

```
% // LDtoCD
% // CDtoBD
% // Tsimplify
```

```
Out[*]=
```

$$\mathcal{L}_V W_{ab}$$

```
Out[*]=
```

$$\nabla_c W_{ab} V^c + \nabla_b V^c W_{ac} + \nabla_a V^c W_{cb}$$

```
Out[*]=
```

$$\partial_c W_{ab} V^c + \partial_b V^c W_{ac} + \Gamma_{bc}^d V^c W_{ad} - \Gamma_{cb}^d V^c W_{ad} + \partial_a V^c W_{cb} + \Gamma_{ac}^d V^c W_{db} - \Gamma_{ca}^d V^c W_{db}$$

```
Out[*]=
```

$$\partial_c W_{ab} V^c + \partial_b V^c W_{ac} - \partial_a V^c W_{bc}$$

LDtoXDRule[]:

$$\mathcal{L}_V w = i_V(dw) + d(i_V w)$$

```
In[*]:= lie /. LDtoXDRule[]
```

```
FtoC[%, {la, lb}]
```

```
Out[*]=
```

$$\mathcal{L}_V \mathbf{dw} + \mathbf{d} \mathcal{L}_V \mathbf{w}$$

```
Out[*]=
```

$$\nabla_c W_{ab} V^c + \nabla_b V^c W_{ac} - \nabla_a V^c W_{bc}$$

Lie 도함수의 좌표 표현:

```
In[*]:= SetDimension[3]
```

```
In[*]:= Fdefine[X[ua], 0]
```

```
In[*]:= 1/2! FtoC[lie, {la, lb}] \times XD[X[ua]] \wedge XD[X[ub]]
```

```
% // SumDum
% // TindexSort
```

```
Out[*]=
```

$$\frac{1}{2} \mathcal{L}_V W_{ab} \mathbf{dX}^a \wedge \mathbf{dX}^b$$

```
Out[*]=
```

$$-\frac{1}{2} \mathcal{L}_V W_{31} \mathbf{dr} \wedge \mathbf{dz} + \frac{1}{2} \mathcal{L}_V W_{13} \mathbf{dr} \wedge \mathbf{dz} - \frac{1}{2} \mathcal{L}_V W_{21} \mathbf{dr} \wedge \mathbf{d\theta} + \frac{1}{2} \mathcal{L}_V W_{12} \mathbf{dr} \wedge \mathbf{d\theta} + \frac{1}{2} \mathcal{L}_V W_{32} \mathbf{dz} \wedge \mathbf{d\theta} - \frac{1}{2} \mathcal{L}_V W_{23} \mathbf{dz} \wedge \mathbf{d\theta}$$

```
Out[*]=
```

$$\mathcal{L}_V W_{13} \mathbf{dr} \wedge \mathbf{dz} + \mathcal{L}_V W_{12} \mathbf{dr} \wedge \mathbf{d\theta} - \mathcal{L}_V W_{23} \mathbf{dz} \wedge \mathbf{d\theta}$$

In[*]:= **CoordRep**[lie, {r, θ, z}]

Out[*]= $\mathcal{L}_V w_{13} \mathbf{dr} \wedge \mathbf{dz} + \mathcal{L}_V w_{12} \mathbf{dr} \wedge \mathbf{d}\theta - \mathcal{L}_V w_{23} \mathbf{dz} \wedge \mathbf{d}\theta$

In[*]:= **CoordRep**[lie]

Out[*]= $\mathcal{L}_V w_{12} \mathbf{dx}^1 \wedge \mathbf{dx}^2 + \mathcal{L}_V w_{13} \mathbf{dx}^1 \wedge \mathbf{dx}^3 + \mathcal{L}_V w_{23} \mathbf{dx}^2 \wedge \mathbf{dx}^3$

In[*]:= **ClearDimension**[] (* default *)

외적의 Lie 도함수:

In[*]:= **w** $\wedge$ **w**
LD[v, %]

Out[*]= **w** $\wedge$ **w**

Out[*]= $2 \mathbf{w} \wedge \mathcal{L}_V \mathbf{w}$

5.2.2 The Interior Product

In[*]:= **Tdefine**[v[ua]]; **Fdefine**[w, 2]

벡터 v와 2-form w의 내적:

In[*]:= **IP**[v, w]
FtoC[% , {la}]

Out[*]= $\mathcal{L}_V \mathbf{w}$

Out[*]= $-v^b w_{ab}$

p-form α 와 q-form β 에 대하여

$$i_V(\alpha \wedge \beta) = (i_V \alpha) \wedge \beta + (-1)^p \alpha \wedge (i_V \beta)$$

In[*]:= **IP**[v, w $\wedge$ w]

Out[*]= $2 \mathcal{L}_V \mathbf{w} \wedge \mathbf{w}$

In[*]:= **Fdefine**[η, 1];
IP[v, η $\wedge$ w]

Out[*]= $\mathbf{w} \mathcal{L}_V \eta - \eta \wedge \mathcal{L}_V \mathbf{w}$

```
In[*]:= RemoveIndexedObject[x];
RemoveIndexedObject[η]
```

내적의 좌표 표현:

```
In[*]:= SetDimension[3]
```

```
In[*]:= Fdefine[X[ua], 0];
SetComponents[X[ua], {x, y, z}]
```

```
In[*]:= FtoC[IP[v, w[]], {la}] × XD[X[ua]]
% // SumDum
% // TindexSort
% // CollectForm
```

Out[*]=

$$-v^a w_{ba} \mathbf{d}x^b$$

Out[*]=

$$-v^3 w_{13} \mathbf{d}x - v^2 w_{12} \mathbf{d}x - v^1 w_{11} \mathbf{d}x - v^3 w_{23} \mathbf{d}y - \\ v^2 w_{22} \mathbf{d}y - v^1 w_{21} \mathbf{d}y - v^3 w_{33} \mathbf{d}z - v^2 w_{32} \mathbf{d}z - v^1 w_{31} \mathbf{d}z$$

Out[*]=

$$-v^3 w_{13} \mathbf{d}x - v^2 w_{12} \mathbf{d}x - v^3 w_{23} \mathbf{d}y + v^1 w_{12} \mathbf{d}y + v^2 w_{23} \mathbf{d}z + v^1 w_{13} \mathbf{d}z$$

Out[*]=

$$\left(-v^3 w_{13} - v^2 w_{12}\right) \mathbf{d}x + \left(-v^3 w_{23} + v^1 w_{12}\right) \mathbf{d}y + \left(v^2 w_{23} + v^1 w_{13}\right) \mathbf{d}z$$

```
In[*]:= CoordRep[IP[v, w], {x, y, z}]
```

Out[*]=

$$\left(-v^3 w_{13} - v^2 w_{12}\right) \mathbf{d}x + \left(-v^3 w_{23} + v^1 w_{12}\right) \mathbf{d}y + \left(v^2 w_{23} + v^1 w_{13}\right) \mathbf{d}z$$

0-form의 Lie 도함수:

```
In[*]:= Fdefine[f, 0]
```

```
In[*]:= LD[v, f]
FtoC[%, {}]
% // CDtoBD
```

Out[*]=

$$\mathcal{L}_v \mathbf{f}$$

Out[*]=

$$\nabla_a \mathbf{f} v^a$$

Out[*]=

$$\partial_a \mathbf{f} v^a$$

5.2.3 The Hodge Star Operator

p -form에 대한 Dual 연산 (참고. R. M. Wald 문제 2-(a)):

$$(*\alpha)_{b_1 \dots b_{n-p}} = \frac{1}{p!} \alpha^{a_1 \dots a_p} \epsilon_{a_1 \dots a_p b_1 \dots b_{n-p}}$$

$$(* * \alpha) = (-1)^{s+p(n-p)} \alpha$$

여기서 s 는 계량 텐서의 signature (number of negative eigenvalues)이다.

```
In[*]:= SetDimension[4]; SetSig[1] (*Minkowski spacetime*)
```

```
In[*]:= Fdefine[w1, 1]; Fdefine[w2, 2]; Fdefine[w3, 3]; Fdefine[w4, 4]
```

```
In[*]:= HodgeStar /@ {w1, w2, w3, w4}
```

```
Out[*]=
```

```
{*w1, *w2, *w3, *w4}
```

HodgeStar는 p -form을 $(n-p)$ -form으로 바꾼다:

```
In[*]:= DegreeForm /@ %
```

```
Out[*]=
```

```
{3, 2, 1, 0}
```

```
In[*]:= w1Star = FtoC[HodgeStar[w1], {la, lb, lc}]
```

```
Out[*]=
```

$$\epsilon^d_{abc} w1_d$$

```
In[*]:= HodgeStar@HodgeStar@w1
```

```
Out[*]=
```

```
w1
```

HodgeStar의 IndexedTensor 표현:

```
In[*]:= DualStar[w1Star, {ld}]
% /. EpsilonProductRule[]
```

```
Out[*]=
```

$$\frac{1}{6} \epsilon^{abc}_d \epsilon^e_{abc} w1_e$$

```
Out[*]=
```

```
w1_d
```

```
In[*]:= FtoC[HodgeStar@w2, {la, lb}]
```

```
Out[*]=
```

$$\frac{1}{2} \epsilon^{cd}_{ab} w2_{cd}$$

In[*]:= FtoC[HodgeStar@w4, {}]

Out[*]=

$$\frac{1}{24} \epsilon^{abcd} w_{4abcd}$$

In[*]:= XD@HodgeStar@w3
FtoC[%, {la, lb}]

Out[*]=

d*w3

Out[*]=

$$\frac{1}{6} \nabla_b w_{3cde} \epsilon_a^{cde} - \frac{1}{6} \nabla_a w_{3cde} \epsilon_b^{cde}$$

In[*]:= ClearDimension[]

3차원 유클리드 공간으로 설정하자:

In[*]:= SetConstantMetric[{1, 1, 1}]

In[*]:= Fdefine[X[ua], 0]; SetComponents[X[ua], {x, y, z}]

In[*]:= Off[XDtoCDfrag]; On[EvaluateBDFlag]; SetCoordinates[{x, y, z}]

In[*]:= HodgeStar /@ {XD[x], XD[y], XD[z]}

Out[*]=

{*dx, *dy, *dz}

In[*]:= (1/2! FtoC[HodgeStar@XD[x], {la, lb}]) XP[XD[X[ua]], XD[X[ub]]]
% // SumDum
% // TindexSort

Out[*]=

$$\frac{1}{2} \partial_a X \epsilon_{bc}^a \mathbf{dX}^b \wedge \mathbf{dX}^c$$

Out[*]=

$$-\frac{1}{2} \epsilon_{32}^1 \mathbf{dy} \wedge \mathbf{dz} + \frac{1}{2} \epsilon_{23}^1 \mathbf{dy} \wedge \mathbf{dz}$$

Out[*]=

$$\epsilon_{23}^1 \mathbf{dy} \wedge \mathbf{dz}$$

In[*]:= CoordRep[HodgeStar@XD[x], {x, y, z}]

Out[*]=

$$\epsilon_{23}^1 \mathbf{dy} \wedge \mathbf{dz}$$

```
In[*]:= % /. Epsilon[a_, b_, c_] => Epsilon[ToDnIndex[a], b, c]
```

```
Out[*]=
```

```
dy^dz
```

```
In[*]:= 1 / 3! FtoC[HodgeStar[1], {la, lb, lc}] * XP[XD[X[ua]], XD[X[ub]], XD[X[uc]]]
% // SumDum
```

```
Out[*]=
```

```
1
-- εabc dXa ∧ dXb ∧ dXc
6
```

```
Out[*]=
```

```
dx^dy^dz
```

```
In[*]:= CoordRep[HodgeStar[1], {x, y, z}]
```

```
Out[*]=
```

```
dx^dy^dz
```

```
In[*]:= ClearConstantMetric[];
On[XDtoCDfrag];
Off[EvaluateBDFlag];
ClearCoordinates[]
```

5.2.4 The Codifferential Operator

p -form ω 에 대한 공식:

$$\delta\omega = (-1)^{s+p(n+2)} * d * \omega = (-1)^p *^{-1} d * \omega$$

여기서 n 은 시공간 차원, s 는 계량 텐서의 signature이다.

```
In[*]:= SetConstantMetric[{1, 1, 1}] (*Euclid space*)
```

```
In[*]:= Fdefine[w0, 0]; Fdefine[w2, 2]; Fdefine[w3, 3]
```

CoXD는 p -form을 $(p-1)$ -form으로 바꾼다:

```
In[*]:= CoXD[w0]
```

```
Out[*]=
```

```
0
```

```
In[*]:= CoXD[w3]
expr = FtoC[%, {la, lb}]
```

```
Out[*]=
```

```
δw3
```

```
Out[*]=
```

```
-∇cw3cab
```


모든 미분 형식에 대해 $\delta^2 \omega$ 는 0이다.

```
In[*]:= CoXD@CoXD@w3
```

```
Out[*]=
```

0

δw_3 의 좌표 표현:

```
In[*]:= Fdefine[X[ua], 0]
```

```
In[*]:= (1/2! expr) XP[XD[X[ua]], XD[X[ub]]]
% // SumDum
% // TindexSort
```

```
Out[*]=
```

$$-\frac{1}{2} \nabla^a w_{3abc} \mathbf{d}x^b \wedge \mathbf{d}x^c$$

```
Out[*]=
```

$$\begin{aligned} & \frac{1}{2} \nabla^1 w_{3121} \mathbf{d}x \wedge \mathbf{d}y - \frac{1}{2} \nabla^1 w_{3112} \mathbf{d}x \wedge \mathbf{d}y + \frac{1}{2} \nabla^2 w_{3221} \mathbf{d}x \wedge \mathbf{d}y - \frac{1}{2} \nabla^2 w_{3212} \mathbf{d}x \wedge \mathbf{d}y + \frac{1}{2} \nabla^3 w_{3321} \mathbf{d}x \wedge \mathbf{d}y - \\ & \frac{1}{2} \nabla^3 w_{3312} \mathbf{d}x \wedge \mathbf{d}y + \frac{1}{2} \nabla^1 w_{3131} \mathbf{d}x \wedge \mathbf{d}z - \frac{1}{2} \nabla^1 w_{3113} \mathbf{d}x \wedge \mathbf{d}z + \frac{1}{2} \nabla^2 w_{3231} \mathbf{d}x \wedge \mathbf{d}z - \\ & \frac{1}{2} \nabla^2 w_{3213} \mathbf{d}x \wedge \mathbf{d}z + \frac{1}{2} \nabla^3 w_{3331} \mathbf{d}x \wedge \mathbf{d}z - \frac{1}{2} \nabla^3 w_{3313} \mathbf{d}x \wedge \mathbf{d}z + \frac{1}{2} \nabla^1 w_{3132} \mathbf{d}y \wedge \mathbf{d}z - \\ & \frac{1}{2} \nabla^1 w_{3123} \mathbf{d}y \wedge \mathbf{d}z + \frac{1}{2} \nabla^2 w_{3232} \mathbf{d}y \wedge \mathbf{d}z - \frac{1}{2} \nabla^2 w_{3223} \mathbf{d}y \wedge \mathbf{d}z + \frac{1}{2} \nabla^3 w_{3332} \mathbf{d}y \wedge \mathbf{d}z - \frac{1}{2} \nabla^3 w_{3323} \mathbf{d}y \wedge \mathbf{d}z \end{aligned}$$

```
Out[*]=
```

$$-\nabla^3 w_{3123} \mathbf{d}x \wedge \mathbf{d}y + \nabla^2 w_{3123} \mathbf{d}x \wedge \mathbf{d}z - \nabla^1 w_{3123} \mathbf{d}y \wedge \mathbf{d}z$$

```
In[*]:= CoordRep[CoXD[w3], {x, y, z}]
```

```
Out[*]=
```

$$-\nabla^3 w_{3123} \mathbf{d}x \wedge \mathbf{d}y + \nabla^2 w_{3123} \mathbf{d}x \wedge \mathbf{d}z - \nabla^1 w_{3123} \mathbf{d}y \wedge \mathbf{d}z$$

CoXD의 IndexedTensor 표현:

```
In[*]:= CoXD[XD[w2]]
FtoC[%, {la, lb}]
```

```
Out[*]=
```

δw_2

```
Out[*]=
```

$$-\nabla^c \nabla_a w_{2bc} + \nabla^c \nabla_b w_{2ac} - \nabla^c \nabla_c w_{2ab}$$

```
In[*]:= XD[CoXD[w2]]
FtoC[%, {la, lb}]
```

```
Out[*]=
dδw2
```

```
Out[*]=
 $\nabla_a \nabla_c w2_b^c - \nabla_b \nabla_c w2_a^c$ 
```

```
In[*]:= ClearConstantMetric[]
```

5.2.5 The Hodge and Bochner Laplacian Operator

3차원 유클리드 공간의 특성으로 설정하자:

```
In[*]:= SetDimension[3]; SetSig[3]
```

```
In[*]:= Fdefine[w2, 2]
```

일반화 라플라스 연산자(Hodge Laplacian):

$$d\delta + \delta d$$

```
In[*]:= CoXD@XD@w2
expr1 = FtoC[%, {la, lb}]
```

```
Out[*]=
δdw2
```

```
Out[*]=
 $-\nabla^c \nabla_a w2_{bc} + \nabla^c \nabla_b w2_{ac} - \nabla^c \nabla_c w2_{ab}$ 
```

```
In[*]:= XD@CoXD@w2
expr2 = FtoC[%, {la, lb}]
```

```
Out[*]=
dδw2
```

```
Out[*]=
 $\nabla_a \nabla_c w2_b^c - \nabla_b \nabla_c w2_a^c$ 
```

```
In[*]:= expr1 + expr2
PutMetric[%, uc]
% // Absorbg
```

```
Out[*]=
 $\nabla_a \nabla_c w2_b^c - \nabla_b \nabla_c w2_a^c - \nabla^c \nabla_a w2_{bc} + \nabla^c \nabla_b w2_{ac} - \nabla^c \nabla_c w2_{ab}$ 
```

```
Out[*]=
 $\nabla_a \nabla_c w2_{bd} g^{cd} - \nabla_b \nabla_c w2_{ad} g^{cd} - \nabla_c \nabla_a w2_{bd} g^{dc} + \nabla_c \nabla_b w2_{ad} g^{dc} - \nabla_c \nabla_d w2_{ab} g^{dc}$ 
```

```
Out[*]=
 $\nabla_a \nabla^c w2_{bc} - \nabla_b \nabla^c w2_{ac} - \nabla^c \nabla_a w2_{bc} + \nabla^c \nabla_b w2_{ac} - \nabla^c \nabla_c w2_{ab}$ 
```

In[*]:= **lapsum = % + CD[uc, lc, w2[la, lb]]**

Out[*]=

$$\nabla_a \nabla^c w_{bc} - \nabla_b \nabla^c w_{ac} - \nabla^c \nabla_a w_{bc} + \nabla^c \nabla_b w_{ac}$$

In[*]:= **CommuteCD[{lb, uc}, lapsum[[2]]]**

Out[*]=

$$-\nabla^c \nabla_b w_{ac} + R_b^c w_{ac} - R_b^c{}^d w_{dc}$$

In[*]:= **ReplacePart[lapsum, 2 → %]**

Out[*]=

$$\nabla_a \nabla^c w_{bc} - \nabla^c \nabla_a w_{bc} + R_b^c w_{ac} - R_b^c{}^d w_{dc}$$

연산 후의 ResetDummies 함수가 자동적으로 동작하므로 (출력 결과만 보면 동일해 보이는) 아래의 두 명령을 한 개의 명령 lapsum = % // Tsimplify로 줄이면 안된다.

주의: 시작

In[*]:= **% // Tsimplify**

Out[*]=

$$\nabla_a \nabla_c w_{bc}^c - \nabla_c \nabla_a w_{bc}^c + R_{bc} w_{ac}^c - R_{acbd} w^{cd}$$

In[*]:= **lapsum2 = %**

Out[*]=

$$\nabla_a \nabla_c w_{bc}^c - \nabla_c \nabla_a w_{bc}^c + R_{bc} w_{ac}^c - R_{acbd} w^{cd}$$

주의: 끝

In[*]:= **CommuteCD[{lc, la}, lapsum2[[2]]]**

Out[*]=

$$-\nabla_a \nabla_c w_{bc}^c - R_{ac} w_{bc}^c - R_{cab}^d w_d^c$$

In[*]:= **ReplacePart[lapsum2, 2 → %]**

Out[*]=

$$R_{bc} w_{ac}^c - R_{ac} w_{bc}^c - R_{cab}^d w_d^c - R_{acbd} w^{cd}$$

In[*]:= **% // Tsimplify**

Out[*]=

$$R_{bc} w_{ac}^c - R_{ac} w_{bc}^c - 2 R_{acbd} w^{cd}$$

따라서 일반화 라플라스 연산자와 보통의 라플라스 연산자의 합을 2-form에 적용하면 곡률 텐서에 의존하는 항만 남는다.